LECTURE NOTES

Lecture 2: Derivatives and Gradients

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Contents

1 Definitions for this course

- $[a, b] = \{x \in R \mid a \leq x \leq b\}$ closed interval
- $(a, b) = \{x \in R \mid a < x < b\}$ open interval

1.1 Linear combination

- with $v_1, v_2 ..., v_k \in \mathbb{R}^n$
- and $\lambda = [\lambda_1, \lambda_2 ..., \lambda_k]^T \in \mathbb{R}^k$

$$x = \lambda_1 v_1 + ... + \lambda_k v_k = \sum_{i=1}^{k} \lambda_i v_i$$

- Conic combination
- Affine combination
- Convex combination

1.2 Convex set

If $x, y \in S$ and $0 \leq \lambda \leq 1$ then $\lambda x + (1 - \lambda) y \in S$

- This means that for a set and any two point x, y all point in between these points must be inside the set

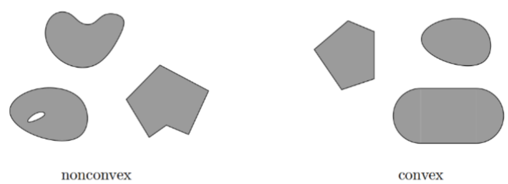

Figure 1: examples of non-convex vs convex

1.3 Convex functions

If for any two points $\forall x, y \in R^n$ with $\alpha \in [0, 1]$ it holds that f(alpha x+(1-alpha)y)>= alpha f(x)+(1-alpha)f(y)

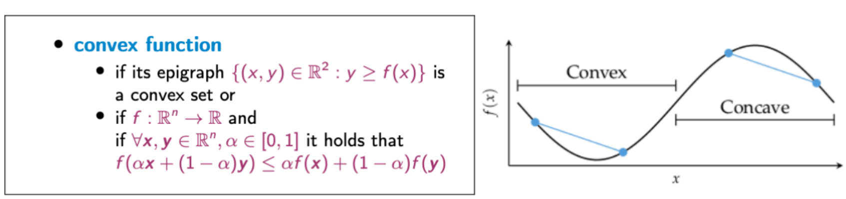

Figure 2: In this example graph is convex on some interval while concave (opposite of convex) in some other interval

1.4 Hulls

For a set of points $S \subseteq \mathbb{R}^n$

- $\text{lin}(S)$ Linear hull (span)
- $\text{cone}(S)$ conic hull
- $\text{aff}(S)$ affine hull

- $\mathrm{conv}(S)$ convex hull

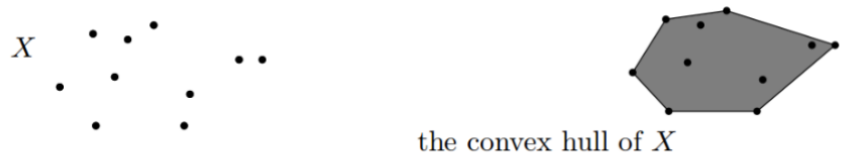

Figure 3: convex hull are the points surrounding all points (much like a rubber band)

$$\mathrm{conv}(X) = \left\{ \lambda_1 x_1 + ... + \lambda_n x_n \mid x_i \in X, \lambda_1, ..., \lambda_n \geq 0 \text{ and} \sum_i \lambda_1 = 1 \right\}$$

2 Derivatives

2.1 Gradient Vector

$$\nabla_S f(x) \equiv \underbrace{\lim_{h \to 0} \frac{f(x+hs) - f(x)}{h}}_{\text{forward difference}} = \underbrace{\lim_{h \to 0} \frac{f\left(x + \frac{hs}{2}\right) - f\left(x - \frac{hs}{2}\right)}{h}}_{\text{central difference}} = \underbrace{\lim_{h \to 0} \frac{f(x) - f(x - hs)}{h}}_{\text{backward difference}}$$

To compute $\nabla_s f(x)$:

- Compute

$$\nabla_s f(x) = \frac{\partial f}{\partial x_1} s_1 + \frac{\partial f}{\partial x_2} s_2 + ... + \frac{\partial f}{\partial x_n} s_n = \nabla f(x)^T S = \nabla f(x) \cdot S$$

2.2 Matrix Calculus

nabla_x b^T x = nabla_x x^T b = b nabla_x x^T A x = (A+A^T)x

2.3 Positive definiteness

Def. A square, symmetric matrix A is **positive definite** if $x^T A x$ is positive for all points other than the origin: $x^T A x > 0$ for all $x \neq 0$.
Def. A square, symmetric matrix A is **positive semidefinite** if $x^T A x$ is always non-negative: $x^T A x \geq 0$ for all x.

Figure 4: A matrix A is positive definite if and only if all its eigenvalues are positive. (This can be used for recognition.)

2.4 LU Decomposition

For P A = L U Where L is a **lower triangular** matrix, U is an **upper triangular** matrix and P is a **permutation matrix** (obtained by rearranging the rows of A)

- Use `LAPACK` (FORTRAN library for python)

3 Symbolic Differentiation

Symbolic derivatives can give valuable insight into the structure of the problem domain and, in some cases, produce analytical solutions of extrema (e.g., solving for $\frac{\partial}{\partial x} f(x) = 0$) that can eliminate the need for derivative calculation altogether.

But they do not lend themselves to efficient runtime calculation of derivative values, as they can get exponentially larger than the expression whose derivative they represent

4 Numerical Differentiation

Neighboring points are used to approximate the derivative

Such as:

$$f(x) \approx \underbrace{\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}}_{\text{forward difference}} = \underbrace{\lim_{h\to 0} \frac{f\left(x+\frac{h}{2}\right) - f\left(x-\frac{h}{2}\right)}{h}}_{\text{central difference}} = \underbrace{\lim_{h\to 0} \frac{f(x) - f(x-h)}{h}}_{\text{backward difference}}$$

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- The O(n) complexity of numerical differentiation for a gradient in n dimensions is the main obstacle to its usefulness in machine learning, where n can be as large as millions or billions in **state-of-the-art** deep learning models

5 Automatic Differentiation

Automatic differentiation techniques are founded on the observation that any function is evaluated by performing a sequence of simple elementary operations involving just one or two arguments at a time:

- addition
- multiplication
- division
- power operation a^b
- trigonometric functions
- exponential functions
- logarithmic $\ln(x)$
- chain rule $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$